

Minimum Cost Pairs

Time limit: 1.00 s

Memory limit: 512 MB

Given an array of n integers, consider pairings of k pairs. Each number can appear in at most one pair, and the cost of a pair (a, b) is $|a - b|$. The cost of a pairing is the sum of all costs of the pairs.

Calculate the minimum costs of pairings for $k = 1, 2, \dots, \lfloor n/2 \rfloor$.

Input

The first line has an integer: the array size.

The next line has n integers $x_1, x_2, \dots, x_n$: the array contents.

Output

Print $\lfloor n/2 \rfloor$ integers: the minimum costs of pairings.

Constraints

- $2 \leq n \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$

Example

Input	Output
8 3 1 2 7 9 3 4 7	0 0 1 6

Explanation: Possible minimum-cost pairings are $[(3, 3)]$, $[(3, 3), (7, 7)]$, $[(1, 2), (3, 3), (7, 7)]$ and $[(1, 2), (3, 3), (4, 7), (7, 9)]$.